QUBO Formulation: Cryptanalysis Prime Factorization Toy Problem

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Problem Statement. Given a composite integer N and a finite set of M candidate factor pairs $\{(p_i, q_i)\}_{i=0}^{M-1}$, the objective is to select exactly one pair whose product equals N. Each candidate is evaluated by a precomputed error metric $e_i = (p_i q_i - N)^2$, potentially augmented with additional penalties for trivial factors. The task is to choose the candidate that minimizes this error while satisfying the selection constraint.

Decision Variables. For each candidate $i \in \{0, 1, \ldots, M-1\}$, introduce a binary variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that the i-th candidate factor pair is selected, and $x_i = 0$ otherwise.

QUBO Derivation. Step 1 — Objective. The original optimization problem is to minimize the total error over the selected candidate:

$$\min_{\mathbf{x} \in \{0,1\}^M} \sum_{i=0}^{M-1} e_i x_i. \tag{1}$$

Step 2 — Constraint. The selection must satisfy the cardinality constraint that exactly one candidate is chosen:

$$\sum_{i=0}^{M-1} x_i = 1. \tag{2}$$

Step 3 — Penalty Term Expansion. We convert the equality constraint into a quadratic penalty term with weight $\lambda > 0$:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{M-1} x_i - 1 \right)^2. \tag{3}$$

Expanding the square yields:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{M-1} x_i^2 - 2 \sum_{i=0}^{M-1} x_i + 1 \right). \tag{4}$$

Applying the idempotent property $x_i^2 = x_i$ for binary variables gives:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{M-1} x_i - 2 \sum_{i=0}^{M-1} x_i + 1 \right) = \lambda \left(1 - \sum_{i=0}^{M-1} x_i \right) = \lambda - \lambda \sum_{i=0}^{M-1} x_i. \tag{5}$$

Thus, the penalty contributes a linear term $-\lambda x_i$ to each variable and a constant offset λ.

Step 4 — Combined Hamiltonian. Adding the objective and the penalty term produces the full QUBO Hamiltonian:

$$H(\mathbf{x}) = \sum_{i=0}^{M-1} e_i x_i + \lambda - \lambda \sum_{i=0}^{M-1} x_i = \sum_{i=0}^{M-1} (e_i - \lambda) x_i + \lambda. \tag{6}$$

Step 5 — Q Matrix and Offset. Reading off the coefficients from the general closed form of $H(\mathbf{x})$, the QUBO parameters are:

$$Q_{i,i} = e_i - \lambda \quad \text{for } i \in \{0, \ldots, M-1\}, \tag{7}$$

$$Q_{i,j} = 0 \quad \text{for } i \neq j, \tag{8}$$

$$c = \lambda. \tag{9}$$

Penalty Weight Justification. Let $E_{\max} = \max_i e_i$. To guarantee that any feasible solution (exactly one $x_i = 1$) has strictly lower energy than any infeasible solution (zero or multiple $x_i = 1$), we require $\lambda > E_{\max}$. If zero variables are selected, the penalty evaluates to λ. If exactly one variable x_i is selected, the energy is e_i. Since $e_i \leq E_{\max} < \lambda$, all feasible solutions are energetically favored over the zero-selection case. If two or more variables are selected, the penalty term grows quadratically, yielding an energy of at least λ, which again exceeds any feasible energy. Therefore, $\lambda > E_{\max}$ ensures that constraint violations are strictly penalized.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^M$ recovers the optimal factor pair because the penalty term enforces the cardinality constraint. Any feasible assignment has energy equal to its objective value, which is bounded above by $E_{\max} < \lambda$. Any infeasible assignment incurs a penalty of at least λ, making its energy strictly greater than that of any feasible assignment. Consequently, the global minimum of H must correspond to a feasible solution, and among those, it selects the one with the smallest error e_i.

Illustrative Example. Consider an instance with $M = 3$ candidates and precomputed errors $e_0 = 4$, $e_1 = 9$, and $e_2 = 1$. We choose $\lambda = 10$, which satisfies $\lambda > \max\{4, 9, 1\}$. Substituting these values into the general formulas yields the diagonal Q matrix entries $Q_{0,0} = 4 - 10 = -6$, $Q_{1,1} = 9 - 10 = -1$, and $Q_{2,2} = 1 - 10 = -9$, with $Q_{i,j} = 0$ for $i \neq j$. The offset is $c = 10$. The concrete Hamiltonian is:

$$H(x_0, x_1, x_2) = -6x_0 - x_1 - 9x_2 + 10. \tag{10}$$

Enumerating the feasible assignments:

$$H(1,0,0) = 4, \quad H(0,1,0) = 9, \quad H(0,0,1) = 1. \tag{11}$$

Infeasible assignments (e.g., $x = (1,1,0)$ or $x = (0,0,0)$) yield energies ≥ 10. The global minimum occurs at $x_2 = 1$ with energy 1, correctly identifying the second candidate as the optimal factor pair.